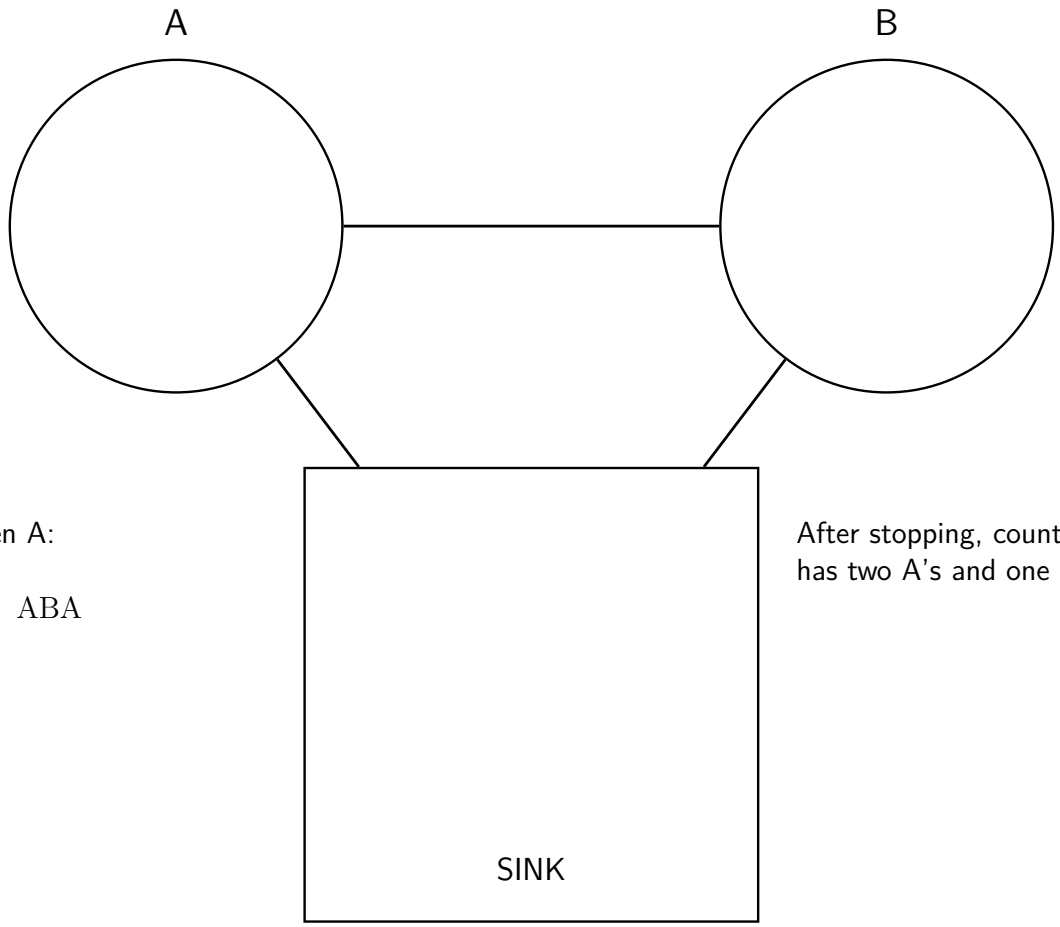
A circle needs one chip for every line touching it before it can share. To share, send one chip along each line. The sink keeps its chips and never shares. Choose one circle at a time. Stop when no circle can share. Empty the sink before each new start. Write each sharing letter in order, then the final piles.

Problem 1: Finish each start in two different orders, if possible. After stopping, count each letter in the sharing word. For the same start, can the final piles agree while the sharing counts differ?



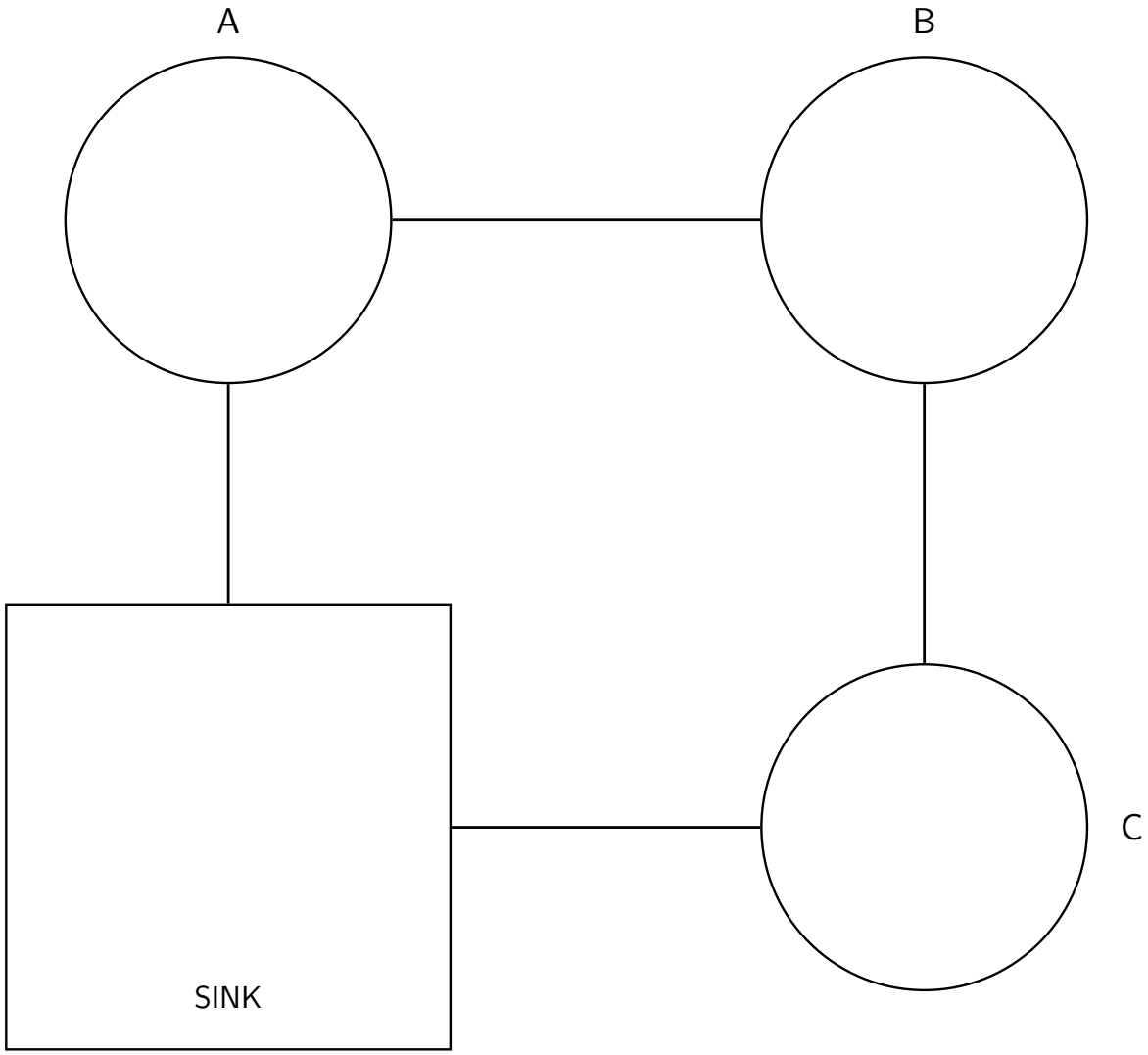
A, then B, then A:

$\mathrm{A} \rightarrow \mathrm{AB} \rightarrow \mathrm{ABA}$

After stopping, count letters: ABA has two A's and one B.

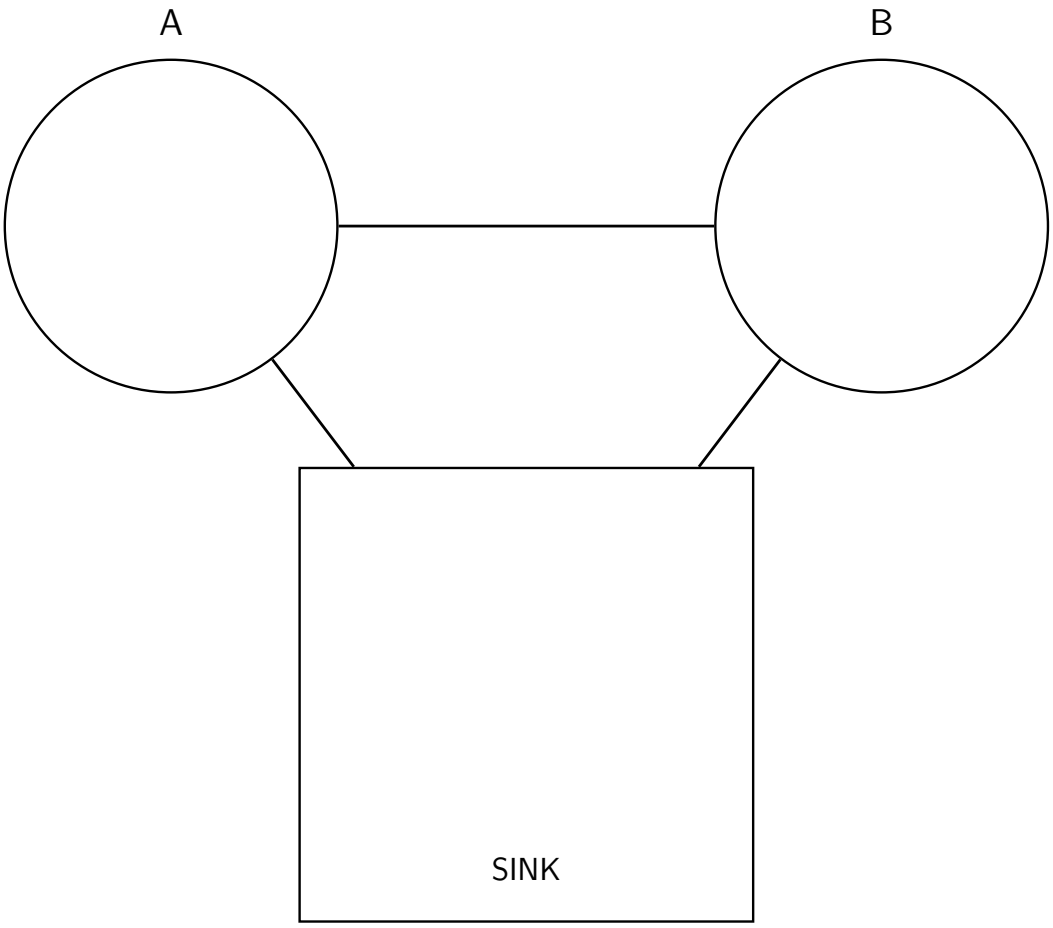
Start A	Start B	Sharing word	Finish A	Finish B
2	2			
2	2			
4	0			
4	0			
3	3			
3	3			
5	4			
5	4			

Problem 2: Start empty. For each pair of batches, compare adding them together with finishing one before adding the other. Try both batch orders. Does the timing of additions change the finish?



First batch	Second batch	Timing	Finish A	Finish B	Finish C
4 at B	2 at A	together			
4 at B	2 at A	first, then second			
4 at B	2 at A	second, then first			
3 at A	3 at C	together			
3 at A	3 at C	first, then second			
3 at A	3 at C	second, then first			

Problem 3: From each start, repeatedly add one chip to A and finish after each addition. Record the pairs you get. Which starts return to themselves? Can two different starts give the same pair after one addition?



Start A	Start B	After 1	After 2	After 3	After 4
0	0				
1	0				
0	1				
1	1				